

Propagation of Errors

1. Introduction

In any analytical measurement, **errors or uncertainties are unavoidable**. When calculations involve more than one measured quantity—such as **addition, subtraction**, multiplication, division, or more complex functions—the **uncertainties associated with each measurement** combine and **propagate**, influencing the final result.

Understanding **how errors propagate** allows chemists and scientists to:

- Assess the **overall uncertainty** in derived results.
- Determine the **reliability and confidence** in conclusions.
- Identify the **major sources of uncertainty** and reduce them effectively.
- Report data with **appropriate precision and accuracy**.

2. Types of Errors Recap

- **Random errors:** Statistical fluctuations caused by unpredictable variations in the measurement process.
- **Systematic errors:** Consistent, reproducible errors caused by faulty equipment or biased techniques.

Error propagation typically **deals with random errors**, as these combine statistically.

3. Understanding Propagation of Errors

What is Propagation of Errors?

- It is the process by which **uncertainties in measured variables** affect the **uncertainty in a calculated result**.

- When a result R depends on several variables $x, y, z, \dots$ each with their own uncertainties, the total uncertainty in R depends on the uncertainties in all these variables.

Mathematical Representation

If:

$$R = f(x, y, z, \dots)$$

where f is some mathematical function of measured quantities, then the **uncertainty in R** (ΔR) is derived from the uncertainties in $x, y, z, \dots$ denoted $\Delta x, \Delta y, \Delta z, \dots$

Using calculus, the **combined uncertainty** is approximately:

$$(\Delta R)^2 = \left(\frac{\partial R}{\partial x} \Delta x \right)^2 + \left(\frac{\partial R}{\partial y} \Delta y \right)^2 + \left(\frac{\partial R}{\partial z} \Delta z \right)^2 + \dots$$

Where:

- $\frac{\partial R}{\partial x}$ is the partial derivative of R with respect to x , indicating **how sensitive R is to changes in x** .
- This equation assumes the errors are **independent and random** (no correlation).

4. Propagation Rules for Common Mathematical Operations

A. Addition and Subtraction

If:

$$R = x + y \quad \text{or} \quad R = x - y$$

Then, the **absolute uncertainties add in quadrature**:

$$\Delta R = \sqrt{(\Delta x)^2 + (\Delta y)^2}$$

- Even when subtracting, uncertainties combine positively.
- Absolute errors are used here.

B. Multiplication and Division

If:

$$R = xy \quad \text{or} \quad R = \frac{x}{y}$$

Then, the **relative (fractional) uncertainties** add in quadrature:

$$\frac{\Delta R}{R} = \sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta y}{y}\right)^2}$$

- This means that the **percentage uncertainties** add up.

C. Powers and Roots

If:

$$R = x^n$$

Then, the **relative uncertainty** in R is:

$$\frac{\Delta R}{R} = |n| \times \frac{\Delta x}{x}$$

D. General Functions

For any function $R = f(x, y, z, \dots)$, use:

$$\Delta R = \sqrt{\left(\frac{\partial R}{\partial x} \Delta x\right)^2 + \left(\frac{\partial R}{\partial y} \Delta y\right)^2 + \dots}$$

5. Examples of Error Propagation

Example 1: Addition

Given:

$$x = 5.0 \pm 0.1, \quad y = 3.0 \pm 0.2$$

Calculate:

$$R = x + y = 8.0$$

Uncertainty:

$$\Delta R = \sqrt{(0.1)^2 + (0.2)^2} = \sqrt{0.01 + 0.04} = \sqrt{0.05} = 0.224$$

Result:

$$R = 8.0 \pm 0.22$$

Example 2: Multiplication

Given:

$$x = 5.0 \pm 0.1, \quad y = 3.0 \pm 0.2$$

Calculate:

$$R = xy = 15.0$$

Relative uncertainties:

$$\frac{\Delta x}{x} = \frac{0.1}{5.0} = 0.02, \quad \frac{\Delta y}{y} = \frac{0.2}{3.0} = 0.067$$

Relative uncertainty in R :

$$\frac{\Delta R}{R} = \sqrt{(0.02)^2 + (0.067)^2} = \sqrt{0.0004 + 0.0045} = \sqrt{0.0049} = 0.07$$

Absolute uncertainty:

$$\Delta R = 0.07 \times 15.0 = 1.05$$

Result:

$$R = 15.0 \pm 1.1$$

6. Techniques to Manage Errors

A. Careful Experimental Design

- Use **precise and calibrated instruments**.
- Reduce the number of variables where possible.
- Choose **methods with higher sensitivity and lower background noise**.
- Increase sample size or replicate measurements to reduce random errors.

B. Calibration and Standardization

- Calibrate instruments regularly.
- Use standards of known concentration to correct systematic errors.
- Apply **blank corrections** to remove background signals.

C. Statistical Analysis

- Perform multiple **measurements** and use **averaging** to reduce random errors.
- Calculate **standard deviations** and **confidence intervals** to quantify uncertainty.
- Apply appropriate **statistical tests** to detect and manage outliers.

D. Minimizing Error Propagation

- Avoid unnecessary complex calculations.
- Use **direct measurements** instead of derived quantities when possible.
- Optimize experimental conditions to improve measurement precision.
- Use error propagation formulas to **identify which variables contribute most to uncertainty** and focus on improving those.

E. Reporting and Interpretation

- Report results with **appropriate significant figures** reflecting uncertainties.
- Include **uncertainty estimates** alongside measurements.
- Interpret data within the **context of the uncertainty** to avoid over-interpretation.

7. Summary

Operation	Formula for Uncertainty	Type of Uncertainty
Addition/Subtraction	$\Delta R = \sqrt{(\Delta x)^2 + (\Delta y)^2}$	Absolute uncertainty
Multiplication/Division	$\frac{\Delta R}{R} = \sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta y}{y}\right)^2}$	Relative uncertainty
Power ($R = x^n$)	$\left(\frac{\Delta R}{R}\right) = n \left(\frac{\Delta x}{x}\right)$	n

8. Conclusion

Understanding and managing **propagation of errors** is crucial in analytical chemistry to ensure **reliable, accurate, and meaningful results**. By using proper error propagation methods, analysts can identify the **most significant sources of uncertainty**, improve experimental design, and confidently communicate the **quality** of their data.